

BB84 Protocol

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I approve



A little introduction

1

Shannon, 1949

Seeing the encrypted message tells you nothing about the original data.

2

Diffie-Hellman

Security relies on computational hardness.
Practical, but not unconditionally secure.

3

Shor's algorithm, 1994

Solves factoring in polynomial time on a quantum computer,
Breaks classical public-key cryptography.

BB84

Bennett & Brassard — 1984

No-cloning theorem

Any eavesdropper necessarily disturbs the quantum channel.

Unconditional security

Guaranteed by physics, not computational assumptions.

The Protocol

1 Qubit Preparation — Alice

Alice picks a random bit (0/1) and a random basis, (rectilinear or diagonal) for each qubit. She encodes the bit as a photon polarization and sends it to Bob.

2 Measurement — Bob

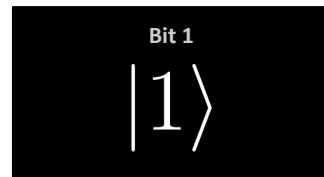
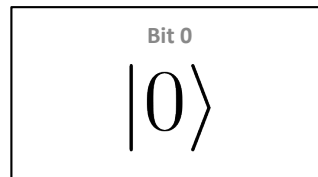
Bob randomly picks a basis for each photon. If his basis matches Alice's, he gets the correct bit. Otherwise the result is random (50/50).

3 Sifting

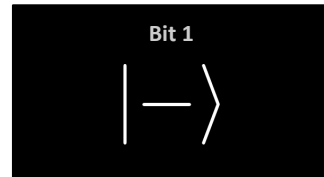
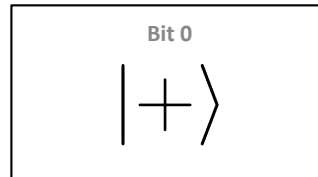
Over a public channel, Alice and Bob compare bases. They discard mismatches and keep the ~50% where bases agreed.

Encoding

Rectilinear $\oplus$



Diagonal $\otimes$



The Protocol

4

Sifting

Alice and Bob publicly compare their bases. They keep only the matching rounds (~50%).

5

Error Estimation

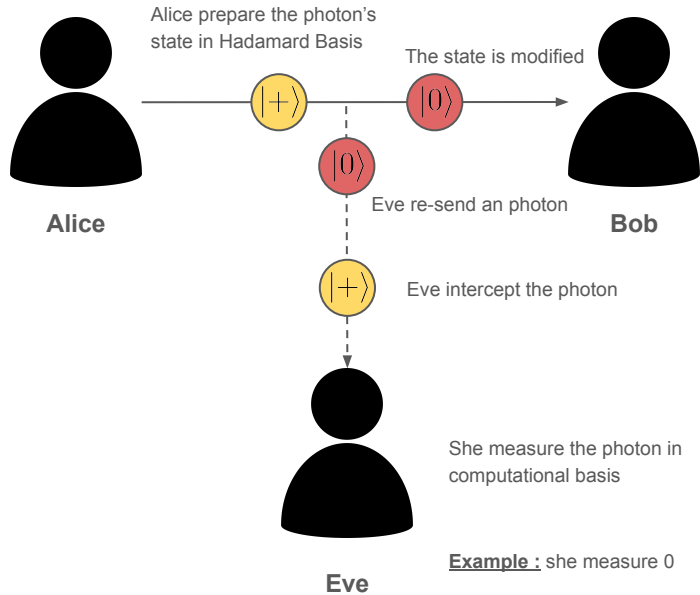
They compare a small subset of bits. If it differs, an eavesdropper is suspected and the key is discarded.

6

The key

The final key is made of the remaining bits after compression.

Security - Eve is a spy 🕶️



Probability of getting the key by chance

$$\begin{aligned} P(\text{Eve guesses one bit}) &= P(\text{right basis}) \cdot P(\text{correct} \mid \text{right basis}) \\ &\quad + P(\text{wrong basis}) \cdot P(\text{correct} \mid \text{wrong basis}) \\ &= \frac{1}{2} \cdot 1 + \frac{1}{2} \cdot \frac{1}{2} = \frac{3}{4} \end{aligned}$$

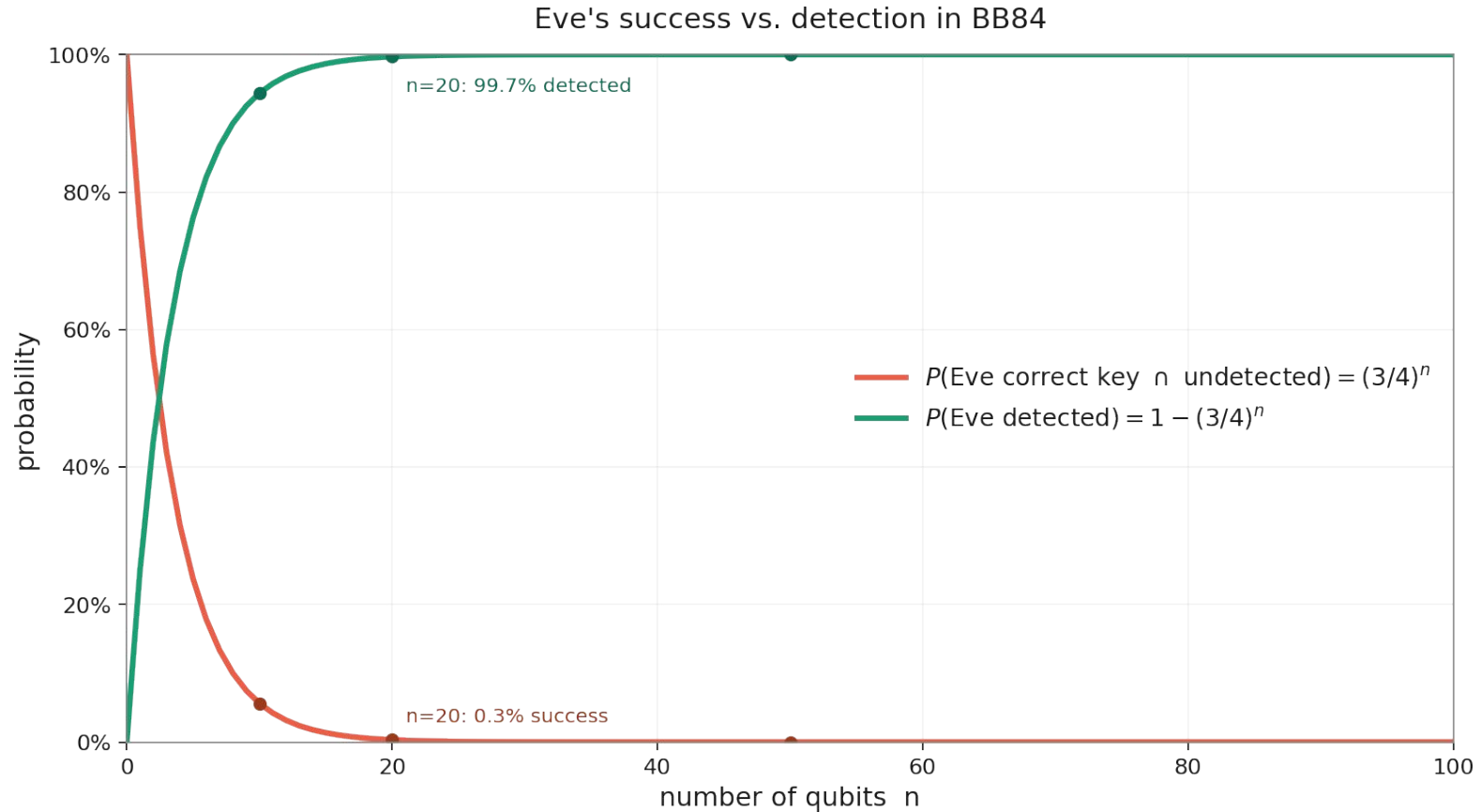
$$P(\text{Eve gets all bits correct}) = \left(\frac{3}{4}\right)^n$$

But with verification...

$$P(\text{Eve undetected}) = (1 - \text{QBER})^m = \left(1 - \frac{1}{4}\right)^m = \left(\frac{3}{4}\right)^m$$

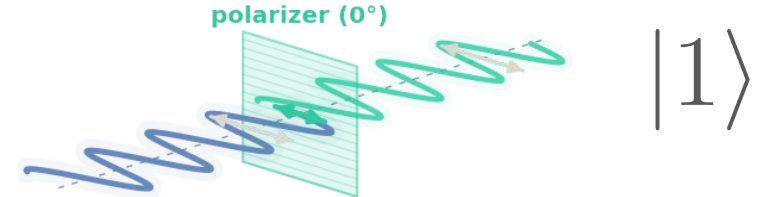
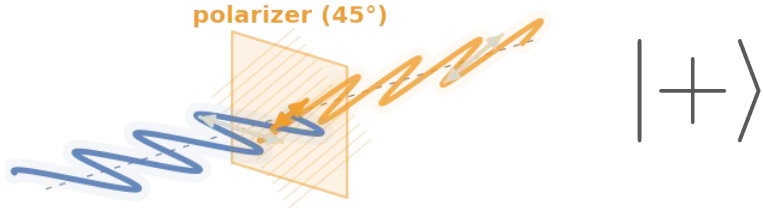
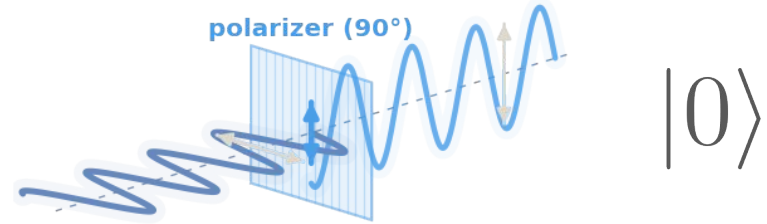
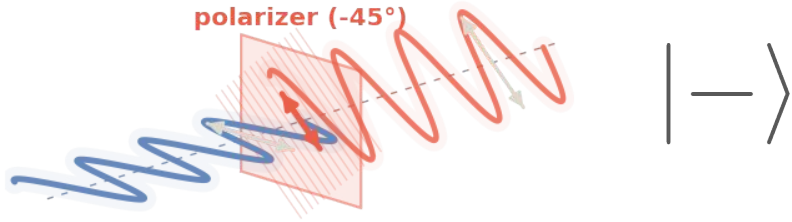
m : number of tested qubits

Security - Eve is a spy 🕵️



Physical Implementation

Using photons



Limits: range, no mature repeaters, imperfect photon sources, vulnerable detectors (blinding, sensitivity, ...)

Our MPQP implementation

Live demo !

Derived protocols

SSP

New Y-basis :

$$|i^+\rangle = \frac{|0\rangle + i|1\rangle}{\sqrt{2}}$$

$$|i^-\rangle = \frac{|0\rangle - i|1\rangle}{\sqrt{2}}$$

- Reduces probability of guessing basis & state
- But security gain is modest

E91

Use of entangled photons in Bell states

- No trust in the device (source AND detectors), both can be controlled by Eve
- Bell inequality violation as the only security guarantee.

Thank you ! Any questions ?

Bibliography

- ❖ Mohamed Elboukhari, Mostafa Azizi et Abdelmalek Azizi, *Analysis of the Security of BB84 by Model Checking*, International Journal of Network Security & Its Applications (IJNSA), Vol. 2, No. 2, April 2010. arXiv:1005.4504.
- ❖ Peter Shor, *Notes 8.370/18.435*, Lecture 31, Fall 2021, MIT.
- ❖ Ramona Wolf, *Quantum Key Distribution: An Introduction with Exercises*, Lecture Notes in Physics, Springer, 2021.